

# A Resource-Constrained Evaluation of Lightweight Adaptation for Cross-Dataset Chest X-ray Report Generation

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## Introduction

- Challenge:** Severe visual and linguistic domain shifts (IU X-Ray → CheXpert Plus) involves substantial differences in both visual characteristics and report-writing styles, making **cross-dataset report generation clinically unreliable**.
- Conventional limits:** Existing lightweight adaptation methods are mainly designed for classification and **may not adequately adapt autoregressive language generation**.
- Motivation:** Investigate whether lightweight adaptation can improve cross-domain report generation under **limited computational resources**, and diagnose why **visual adaptation alone may be insufficient**.

## Our Contribution

- Benchmark** lightweight adaptation strategies for cross-dataset chest X-ray report generation, including **four LoRA-based variants, router-only MoME-TTA, and MetaVIB**.
- Quantify the IU X-Ray-to-CheXpert Plus visual domain gap using **feature-space Mahalanobis distance** and show strong separation in the BiomedGPT visual representation.
- Adapt a **classification-oriented TTA approach** to an **autoregressive report generation** backbone and analyze the limitations of router-only inference-time adaptation.
- Show that small improvements in **lexical metrics** do not necessarily correspond to clinically better reports, highlighting the need for **evaluation protocols that capture omissions, unsupported findings, and target-domain report style**.

## Methods

### Dataset & Experimental Setup

**Source → Target:** IU X-Ray → CheXpert Plus    **Base Model:** BiomedGPT pretrained on IU X-Ray  
Experiments were conducted under a limited-resource Google Colab setting

### Frozen BiomedGPT Backbone

Only the lightweight modules below are trained

#### LoRa-based Adaptation

- Vision-only:** Adapt visual encoder features
- Vision-decoder:** Adapt visual features + updates decoder cross-attention
- KD-ASL:** Combine CE training with token-level knowledge distillation and pathology-level prediction
- SWAD-inspired:** Average low-validation-loss LoRA checkpoints for improved robustness

#### MoME-TTA

- Unlabeled** target-domain adaptation method.
- Freeze experts, and language modules, optimizing only an **entropy-based router** during inference.
- Only **0.009% of model parameters** are trainable.

#### MetaVIB

- Construct pseudo-domains within IU X-Ray using **frontal/lateral views and feature clustering**.
- Evaluates whether **classification-oriented** latent regularization transfers to **autoregressive** report generation.

### Visual Domain Shift Analysis

Mahalanobis Distance

- Extract visual features from the pretrained BiomedGPT encoder.
- Estimate the IU X-Ray feature distribution and measure target-domain distance.
- Use AUROC to quantify source-target separability.

**Table 1.** Mahalanobis distance evaluation between IU X-Ray and CheXpert Plus.

| Metric                      | Value    |
|-----------------------------|----------|
| Source mean (IU X-Ray)      | 22.9422  |
| Target mean (CheXpert Plus) | 165.8608 |
| AUROC                       | 1.0000   |

→ Strong visual domain shift exists before adaptation.

**Table 2.** OOD Detection performance of MetaVIB.

| Metric            | Value  |
|-------------------|--------|
| AUROC             | 0.5867 |
| AUPR              | 0.5783 |
| FPR95             | 0.9283 |
| ID mean distance  | 2.5468 |
| OOD mean distance | 3.2550 |

→ There is a substantial overlap between the pseudo-domains. This suggests that view-based pseudo-OOD construction provides only weak supervision for learning domain-robust representations.

→ We provide a controlled diagnostic study of what currently fails and why, by comparing **lightweight adaptation strategies** for an IU X-Ray-pretrained BiomedGPT model transferred to CheXpert Plus.

## Results

### Quantitative Evaluation

Generated reports are evaluated using NLG (Natural Language Generation) metrics such as ROUGE and BLEU.

**Table 2.** Quantitative comparison of BiomedGPT models with different adaptation strategies on the CheXpert Plus dataset. Bold values denote the best point estimate for each metric.

| Model                 | ROUGE-1       | ROUGE-L       | BLEU-1        | BLEU-4        | BERTScore F1  |
|-----------------------|---------------|---------------|---------------|---------------|---------------|
| Source-only BiomedGPT | 0.2213        | 0.1560        | 0.0993        | 0.0158        | 0.7972        |
| Vision-only LoRA      | 0.2233        | <b>0.1578</b> | 0.0999        | 0.0149        | 0.7971        |
| Vision-decoder LoRA   | 0.2022        | 0.1450        | 0.0859        | 0.0127        | 0.7984        |
| KD-ASL LoRA           | 0.2094        | 0.1510        | 0.0917        | 0.0146        | 0.7973        |
| SWAD-inspired LoRA    | 0.1995        | 0.1447        | 0.0832        | 0.0127        | <b>0.7986</b> |
| MoME-TTA              | 0.2221        | 0.1567        | 0.1001        | 0.0166        | 0.7973        |
| MetaVIB               | <b>0.2248</b> | 0.1567        | <b>0.1029</b> | <b>0.0167</b> | 0.7977        |

→ No lightweight adaptation method consistently improved report generation under severe domain shift.

### Overall Findings

- A **severe visual domain gap** was observed between IU X-Ray and CheXpert Plus with **Mahalanobis AUROC reaching 1.00**.
- Nevertheless, **LoRA, MoME-TTA, and MetaVIB produced only marginal improvements**, suggesting that lightweight feature-level adaptation alone is insufficient under severe domain shift.
- Visual alignment does not guarantee clinical adaptation**, as target-domain reporting style and decoder-level clinical semantics remain largely unaddressed.
- NLG metrics do not fully reflect clinical quality**, highlighting a mismatch between lexical similarity and clinically faithful reporting.
- These results suggest that **classification-oriented adaptation objectives are not directly transferable to autoregressive medical report generation**, where decoder-level language and clinical semantics are critical.
- Future work should jointly address **visual robustness, target-domain language, and decoder-level clinical semantics**, while validating the performance-resource trade-off against full-model fine-tuning.

### Exploratory Qualitative Evaluation

We conducted LLM-assisted review of **50 cases** focusing on clinically relevant error patterns.

- Vision-only LoRA:** favorable lexical scores, but generated reports were often incomplete or generic.
- Vision-decoder / SWAD-LoRA:** lower lexical scores, but sometimes produced more clinically specific reports.
- KD-ASL LoRA:** reduced some unsupported findings, but improvements were inconsistent.
- MoME-TTA & MetaVIB:** showed limited qualitative differences from the source-only model.

**Table S2.** Change in qualitative scores relative to baseline ( $\Delta$  = method – baseline). Positive values indicate improvement over baseline. For omission and normal template bias, positive  $\Delta$  denotes fewer omissions or less bias relative to baseline.

| Model               | $\Delta$ Semantic | $\Delta$ Omission | $\Delta$ Major omis. | $\Delta$ Normal bias | $\Delta$ Specificity |
|---------------------|-------------------|-------------------|----------------------|----------------------|----------------------|
| Vision-decoder LoRA | +0.06             | +0.08             | +0.08                | +0.10                | +0.02                |
| SWAD-inspired LoRA  | +0.04             | +0.10             | +0.06                | +0.10                | -0.02                |
| KD-ASL LoRA         | -0.10             | -0.04             | -0.02                | -0.08                | -0.10                |
| Vision-only LoRA    | -0.18             | -0.10             | -0.06                | -0.12                | -0.06                |
| MoME-TTA            | -0.02             | -0.02             | 0.00                 | 0.00                 | 0.00                 |
| MetaVIB             | -0.02             | -0.06             | -0.04                | +0.08                | 0.00                 |

→ Standard NLG metrics did not consistently reflect clinical report quality.